

Linux Commands

1. Create user with name Techie and provide sudo access to user.

```
asif@asif-virtual-machine:~$ sudo adduser techie
Adding user `techie' ...
Adding new group `techie' (1002) ...
Adding new user `techie' (1002) with group `techie' ...
Creating home directory `/home/techie' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for techie
Enter the new value, or press ENTER for the default
    Full Name []: techi
    Room Number []: 401
    Work Phone []: hgfd
    Home Phone []: hgfd
    Other []: hdd
Is the information correct? [Y/n] y
asif@asif-virtual-machine:~$ echo $?
0
asif@asif-virtual-machine:~$ sudo usermod -aG sudo techie
asif@asif-virtual-machine:~$ echo $?
0
asif@asif-virtual-machine:~$
```

Purpose: To understand linux user management and sudo access.

Objective: Create a new user account and grant administrative privileges.

Implementation: Created the user techie using adduser and added the user to the sudo group using usermod.

Conclusion: The user was successfully created and granted administrative privileges.

2. Navigate to the home directory.

```
asif@asif-virtual-machine:~$ cd ~
asif@asif-virtual-machine:~$ echo $?
0
asif@asif-virtual-machine:~$
```

Purpose: To learn directory navigation.

Objective: Access to the user's home directory.

Implementation: Used the cd ~ command.

Conclusion: Successfully navigated to the home directory.

3. Create a new directory.

```
asif@asif-virtual-machine:~$ mkdir swithboard
asif@asif-virtual-machine:~$ echo $?
0
```

Purpose: To learn directory creation.

Objective: Create a new folder.

Implementation: Used the mkdir command.

Conclusion: A new directory was created successfully.

4. List the contents of a directory.

```
asif@asif-virtual-machine:~$ ls -la
total 88
drwxr-x--- 15 asif asif 4096 Aug  7 15:30 .
drwxr-x---  5 root root 4096 Aug  7 15:20 ..
-rw-r--r--  1 asif asif  298 Aug  6 17:48 archive.zip
-rw-r--r--  1 asif asif    0 Aug  6 17:46 asif
-rw-r--r--  1 asif asif  356 Aug  6 16:48 .bash_history
-rw-r--r--  1 asif asif  220 Aug  6 12:04 .bash_logout
-rw-r--r--  1 asif asif 3771 Aug  6 12:04 .bashrc
drwxr-x--- 12 asif asif 4096 Aug  6 15:23 .cache
drwxr-x--- 11 asif asif 4096 Aug  6 15:22 .config
drwxr-xr-x  2 asif asif 4096 Aug  6 12:19 Desktop
drwxr-xr-x  2 asif asif 4096 Aug  6 12:19 Documents
drwxr-xr-x  2 asif asif 4096 Aug  6 12:19 Downloads
-rw-r--r--  1 asif asif    9 Aug  6 18:08 gaming
-rw-r--r--  1 asif asif  164 Aug  6 17:16 hello.gz.bz2.xz
-rw-r--r--  1 asif asif    0 Aug  6 17:46 lenovo
drwxr-x---  3 asif asif 4096 Aug  6 12:19 .local
drwxr-xr-x  2 asif asif 4096 Aug  6 12:19 Music
drwxr-xr-x  3 asif asif 4096 Aug  6 15:11 Pictures
-rw-r--r--  1 asif asif  807 Aug  6 12:04 .profile
drwxr-xr-x  2 asif asif 4096 Aug  6 12:19 Public
drwxr-x---  3 asif asif 4096 Aug  6 12:19 snap
-rw-r--r--  1 asif asif    0 Aug  6 16:53 .sudo_as_admin_successful
drwxrwxr-x  2 asif asif 4096 Aug  7 15:30 swithboard
drwxr-xr-x  2 asif asif 4096 Aug  6 12:19 Templates
drwxr-xr-x  2 asif asif 4096 Aug  6 12:19 Videos
asif@asif-virtual-machine:~$ echo $?
0
```

Purpose: To learn how to inspect directory contents.

Objective: Display files and folders.

Implementation: Used the ls -la command.

Conclusion: Successfully displayed all directory contents.

5. Change the current directory.

```
asif@asif-virtual-machine:~$ cd swithboard
asif@asif-virtual-machine:~/swithboard$ echo $?
0
```

Purpose: To understand directory navigation.

Objective: Switch to another directory.

Implementation: Used the cd command.

Conclusion: Successfully changed the current working directory.

6. Create a new empty file.

```
asif@asif-virtual-machine:~/swithboard$ touch charger
asif@asif-virtual-machine:~/swithboard$ echo $?
0
```

Purpose: To learn file creation.

Objective: Create an empty file.

Implementation: Used the touch command.

Conclusion: Successfully created an empty file.

7. View the contents of a file.

```
asif@asif-virtual-machine:~/swithboard$ cat charger
asif@asif-virtual-machine:~/swithboard$ echo $?
0
```

Purpose: To understand file viewing.

Objective: Display file contents.

Implementation: Used the cat command.

Conclusion: Successfully viewed the file contents.

8. Copy a file to another location.

```
asif@asif-virtual-machine:~/swithboard$ cp charger ~/charging
asif@asif-virtual-machine:~/swithboard$ echo $?
0
```

Purpose: To learn file copying.

Objective: Duplicate a file.

Implementation: Used the cp command.

Conclusion: The file was Successfully copied.

9. Move a file to another location.

```
asif@asif-virtual-machine:~/swithboard$ mv charger ~/
asif@asif-virtual-machine:~/swithboard$ echo $?
0
```

Purpose: To understand file relocation.

Objective: Move a file.

Implementation: Used the mv command.

Conclusion: The file was Successfully moved.

10. Rename a file.

```
asif@asif-virtual-machine:~/swithboard$ mv ~/charger phone
asif@asif-virtual-machine:~/swithboard$ echo $?
0
```

Purpose: To understand file renaming.

Objective: Rename an existing file.

Implementation: Used the mv command with a new filename.

Conclusion: The file was Successfully renamed.

11. Delete a file.

```
asif@asif-virtual-machine:~/swithboard$ rm phone
asif@asif-virtual-machine:~/swithboard$ echo $?
0
```

Purpose: To understand file deletion.

Objective: Remove a file.

Implementation: Used the rm command.

Conclusion: The file was Successfully deleted.

12. View the current date and time.

```
asif@asif-virtual-machine:~$ date  
Friday 07 August 2026 03:48:54 PM IST
```

Purpose: To learn basic system information command.

Objective: Display the system date and time.

Implementation: Used the date command.

Conclusion: The current system date and time were displayed successfully.

13. Check the system uptime.

```
asif@asif-virtual-machine:~$ uptime  
15:49:12 up 9:43, 1 user, load average: 0.01, 0.01, 0.00
```

Purpose: To monitor system availability.

Objective: View how longb the system has been running.

Implementation: Used the uptime command.

Conclusion: The system uptime information was displayed successfully.

14. View the running processes.

```
asif@asif-virtual-machine:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1        0  0 06:05 ?        00:00:17 /sbin/init auto noprompt splash
root           2        0  0 06:05 ?        00:00:00 [kthreadd]
Files         3        2  0 06:05 ?        00:00:00 [pool_workqueue_release]
root           4        2  0 06:05 ?        00:00:00 [kworker/R-rcu_g]
root           5        2  0 06:05 ?        00:00:00 [kworker/R-rcu_p]
root           6        2  0 06:05 ?        00:00:00 [kworker/R-slub_]
root           7        2  0 06:05 ?        00:00:00 [kworker/R-netns]
root          12        2  0 06:05 ?        00:00:00 [kworker/R-mm_pg]
root          13        2  0 06:05 ?        00:00:00 [rcu_tasks_kthread]
root          14        2  0 06:05 ?        00:00:00 [rcu_tasks_rude_kthread]
root          15        2  0 06:05 ?        00:00:00 [rcu_tasks_trace_kthread]
root          16        2  0 06:05 ?        00:00:00 [ksoftirqd/0]
root          17        2  0 06:05 ?        00:00:05 [rcu_preempt]
root          18        2  0 06:05 ?        00:00:00 [migration/0]
root          19        2  0 06:05 ?        00:00:00 [idle_inject/0]
root          20        2  0 06:05 ?        00:00:00 [cpuhp/0]
root          21        2  0 06:05 ?        00:00:00 [cpuhp/1]
root          22        2  0 06:05 ?        00:00:00 [idle_inject/1]
root          23        2  0 06:05 ?        00:00:01 [migration/1]
root          24        2  0 06:05 ?        00:00:00 [ksoftirqd/1]
root          29        2  0 06:05 ?        00:00:00 [kdevtmpfs]
root          30        2  0 06:05 ?        00:00:00 [kworker/R-inet_]
root          31        2  0 06:05 ?        00:00:00 [kauditd]
root          33        2  0 06:05 ?        00:00:01 [khungtaskd]
root          34        2  0 06:05 ?        00:00:00 [oom_reaper]
root          37        2  0 06:05 ?        00:00:00 [kworker/R-write]
root          38        2  0 06:05 ?        00:00:03 [kcompactd0]
root          39        2  0 06:05 ?        00:00:00 [ksmd]
root          42        2  0 06:05 ?        00:00:00 [khugepaged]
root          43        2  0 06:05 ?        00:00:00 [kworker/R-kint]
root          44        2  0 06:05 ?        00:00:00 [kworker/R-kbloc]
root          45        2  0 06:05 ?        00:00:00 [kworker/R-blkcg]
root          46        2  0 06:05 ?        00:00:00 [irq/9-acpi]
root          47        2  0 06:05 ?        00:00:00 [kworker/R-tpm_d]
root          48        2  0 06:05 ?        00:00:00 [kworker/R-ata_s]
root          49        2  0 06:05 ?        00:00:00 [kworker/R-md]
root          50        2  0 06:05 ?        00:00:00 [kworker/R-md_bl]
root          51        2  0 06:05 ?        00:00:00 [kworker/R-edac_]
root          52        2  0 06:05 ?        00:00:00 [kworker/R-devfr]
root          53        2  0 06:05 ?        00:00:00 [watchdogd]
root          54        2  0 06:05 ?        00:00:00 [kworker/R-quota]
root          56        2  0 06:05 ?        00:00:00 [kswapd0]
```

Purpose: To monitor system activity.

Objective: Display active processes.

Implementation: Used the `ps -ef` command.

Conclusion: The running process was displayed successfully.

15. Install a package using the package manager (e.g., apt or yum).

```
asif@asif-virtual-machine:~$ sudo apt install
[sudo] password for asif:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugins5 libgcb-1.0-0 libsmbios-c2
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
asif@asif-virtual-machine:~$ echo $?
```

Purpose: To understand package management.

Objective: Install software using the package manager.

Implementation: Used `apt install` command to install a package.

Conclusion: The package was installed successfully.

16. Update the system packages.

```
asif@asif-virtual-machine:~$ sudo apt update
Get:1 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Get:3 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:4 http://security.ubuntu.com/ubuntu jammy-security/main amd64 DEP-11 Metadata [74.3 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:6 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 DEP-11 Metadata [127 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [3,714 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu jammy-updates/main i386 Packages [1,057 kB]
Get:9 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 DEP-11 Metadata [117 kB]
Get:10 http://in.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1,279 kB]
Get:11 http://in.archive.ubuntu.com/ubuntu jammy-updates/universe i386 Packages [815 kB]
Get:12 http://in.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 DEP-11 Metadata [360 kB]
Get:13 http://in.archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 DEP-11 Metadata [940 B]
Get:14 http://in.archive.ubuntu.com/ubuntu jammy-backports/main amd64 DEP-11 Metadata [5,772 B]
Get:15 http://in.archive.ubuntu.com/ubuntu jammy-backports/universe amd64 DEP-11 Metadata [12.4 kB]
Fetched 7,947 kB in 3s (2,453 kB/s)

Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
asif@asif-virtual-machine:~$ echo $?
0
```

Purpose: To maintain system security and stability.

Objective: Update installed software packages.

Implementation: Used apt update command.

Conclusion: The system packages were updated successfully.

17. Create a symbolic link.

```
asif@asif-virtual-machine:~$ ln -s /home/$user/tecnote link_tecnote
asif@asif-virtual-machine:~$ echo $?
0
```

Purpose: To understand linux links.

Objective: Create a symbolic link.

Implementation: Used the ln -s command.

Conclusion: The symbolic link was created successfully.

18. Search for files using the find command.

```
asif@asif-virtual-machine:~$ find /home -name "asif"
find: '/home/techie': Permission denied
find: '/home/shaik': Permission denied
/home/asif
/home/asif/asif
asif@asif-virtual-machine:~$ echo $?
```

Purpose: To learn efficient file searching.

Objective: Locate files in the linux.

Implementation: Used the find command with the required search criteria.

Conclusion: The required file was located successfully using the find command.

19. Grant or revoke permissions on a file or directory.

```
asif@asif-virtual-machine:~$ touch permanent
asif@asif-virtual-machine:~$ chmod +x permanent
asif@asif-virtual-machine:~$ ls -l permanent
-rwxrwxr-x 1 asif asif 0 Aug  7 18:11 permanent
asif@asif-virtual-machine:~$ chmod -x permanent
asif@asif-virtual-machine:~$ echo $?
0
```

Purpose: To understand Linux file security.

Objective: Modify file or directory permissions.

Implementation: Used chmod command.

Conclusion: permissions were updated successfully.

20. Kill a running process.

```
asif@asif-virtual-machine:~$ sudo kill -9 7270
[sudo] password for asif:
```

Purpose: To learn process management.

Objective: Terminate a running process.

Implementation: Identified the process ID and executed kill PID.

Conclusion: The selected process was terminated successfully.

21. Compress and decompress files using tar.

Purpose: Compression is used for backups, transferring files, and reducing storage requirements.

Objective: To compress and extract files and directories using the `tar` command.

Implementation:

```
asif@asif-virtual-machine:~$ tar cvf project.tar project/
project/
asif@asif-virtual-machine:~$ echo $?
0
asif@asif-virtual-machine:~$ tar -czvf project.tar.gz project/
project/
asif@asif-virtual-machine:~$ echo $?
0
asif@asif-virtual-machine:~$ tar -xzvf project.tar.gz
project/
asif@asif-virtual-machine:~$ echo $?
0
```

Conclusion:

`tar -czvf project.tar.gz project/` ----- for compress

`tar -xzvf project.tar.gz`-----for decompress

Tar is useful for backups, transferring folders, and saving disk space.

22. Monitor system resources with `top` or `htop`.

Purpose: System administrators use these commands to identify processes consuming excessive resources.

Objective: To monitor CPU, memory, processes, and system load.

Implementation:

```
asif@asif-virtual-machine:~$ top
top - 10:45:23 up 12:55, 1 user, load average: 0.21, 0.21, 0.11
Tasks: 299 total, 1 running, 298 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.2 sy, 0.0 ni, 99.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7894.5 total, 5124.5 free, 1156.4 used, 1613.6 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 6420.2 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR  S  %CPU  %MEM     TIME+ COMMAND
  587 systemd+ 20   0   14836    6884   6044  S   0.3   0.1   1:15.58 systemd-oomd
 3397 asif      20   0 559464   57800  43892  S   0.3   0.7   0:56.43 gnome-terminal-
 8860 root      20   0      0      0      0  I   0.3   0.0   0:00.23 kworker/1:2-events
    1 root      20   0 167812   13272   8392  S   0.0   0.2   0:25.04 systemd
    2 root      20   0      0      0      0  S   0.0   0.0   0:00.92 kthreadd
    3 root      20   0      0      0      0  S   0.0   0.0   0:00.00 pool_workqueue_release
    4 root      0 -20      0      0      0  I   0.0   0.0   0:00.00 kworker/R-rcu_g
    5 root      0 -20      0      0      0  I   0.0   0.0   0:00.00 kworker/R-rcu_p
    6 root      0 -20      0      0      0  I   0.0   0.0   0:00.00 kworker/R-slab_
    7 root      0 -20      0      0      0  I   0.0   0.0   0:00.00 kworker/R-netns
   12 root      0 -20      0      0      0  I   0.0   0.0   0:00.00 kworker/R-mm_pe
   13 root      20   0      0      0      0  I   0.0   0.0   0:00.00 rcu_tasks_kthread
   14 root      20   0      0      0      0  I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
   15 root      20   0      0      0      0  I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
   16 root      20   0      0      0      0  S   0.0   0.0   0:00.59 ksoftirqd/0
   17 root      20   0      0      0      0  I   0.0   0.0   0:06.85 rcu_preempt
   18 root      rt   0      0      0      0  S   0.0   0.0   0:00.97 migration/0
```

Conclusion:

Use top - when it is already available on the server .

Use htop- when you want a more beginner-friendly, colorful view of CPU, RAM, and running processes.

23. Create and manage user groups.

Purpose: Groups allow multiple users to share access to files and resources.

Objective: To create users and groups and manage their permissions.

Implementation:

```
asif@asif-virtual-machine:~$ sudo groupadd management
asif@asif-virtual-machine:~$ echo $?
0
asif@asif-virtual-machine:~$ sudo usermod -aG management asif
asif@asif-virtual-machine:~$ echo $?
0
asif@asif-virtual-machine:~$ groups asif
asif : asif adm cdrom sudo dip plugdev lpadmin lxd sambashare management
asif@asif-virtual-machine:~$ echo $?
0
asif@asif-virtual-machine:~$ sudo gpasswd -d asif management
Removing user asif from group management
asif@asif-virtual-machine:~$ sudo groupdel -n management
groupdel: invalid option -- 'n'
Usage: groupdel [options] GROUP

Options:
  -h, --help                display this help message and exit
  -R, --root CHROOT_DIR     directory to chroot into
  -P, --prefix PREFIX_DIR   prefix directory where are located the /etc/* files
  -f, --force                delete group even if it is the primary group of a user
  --extrausers               Use the extra users database

asif@asif-virtual-machine:~$ sudo groupdel management
asif@asif-virtual-machine:~$ echo $?
0
```

Conclusion: Groups make access control easier

Sudo groupadd management----created a group named management.

Sudo usermod –aG management asif----- added a user named asif to management group.

Sudo gpasswd –d asif management-----removing user asif from management group

Sudo groupdel management-----deleting management group

24. Set up SSH password less authentication.

Purpose: Passwordless SSH is commonly used in DevOps for servers, automation, CI/CD, and remote administration.

Objective: To configure SSH login using an SSH key instead of a password.

Implementation:

```
asif@asif-virtual-machine:~/Desktop$ cd ~
asif@asif-virtual-machine:~$ ssh-keygen -t ed25519
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/asif/.ssh/id_ed25519): asif
asif already exists.
Overwrite (y/n)? y
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in asif
Your public key has been saved in asif.pub
The key fingerprint is:
SHA256:EMKudeZqE1aLzQujrrPILCyCjnt47S6eJ0t1pdGqQLs asif@asif-virtual-machine
The key's randomart image is:
+--[ED25519 256]--+
|      . . .      |
|      . . .      |
|      o = .      |
|    . o X + S    |
| . o * X         |
|++.= O .        |
| Help .         |
+----[SHA256]-----+
asif@asif-virtual-machine:~$ echo $?
```

Conclusion: Your private key stays on your computer; the public key is added to the server.

25. Monitor log files using tail or grep.

Purpose: Logs are essential for troubleshooting applications, servers, authentication problems, and system errors.

Objective: To monitor and search Linux log files.

Implementation:

```
asif@asif-virtual-machine:~$ tail -f /var/log/syslog
Aug  8 15:45:42 asif-virtual-machine dbus-daemon[1138]: [session uid=1000 pid=1138] Successfully activated service 'org.freedesktop.Tracker3.Miner.Extract'
Aug  8 15:45:42 asif-virtual-machine systemd[1060]: Started Tracker metadata extractor.
Aug  8 15:45:42 asif-virtual-machine tracker-extract[10436]: Task for 'file:///home/asif/asif.pub' finished with error: Could not get any metadata for url:'file:///home/asif/asif.pub' and mime:'application/vnd.ms-publisher'
Aug  8 15:49:35 asif-virtual-machine systemd[1]: Starting Refresh fwupd metadata and update motd...
CDROM 3:49:36 asif-virtual-machine fwupdmgrr[10451]: Updating lvfs
Aug  8 15:49:43 asif-virtual-machine fwupdmgrr[10451]: Successfully downloaded new metadata:
Aug  8 15:49:43 asif-virtual-machine fwupdmgrr[10451]:  • No devices are updatable
Aug  8 15:49:43 asif-virtual-machine systemd[1]: fwupd-refresh.service: Deactivated successfully.
Aug  8 15:49:43 asif-virtual-machine systemd[1]: Finished Refresh fwupd metadata and update motd.
Aug  8 15:49:43 asif-virtual-machine systemd[1]: fwupd-refresh.service: Consumed 2.647s CPU time.
```

Conclusion: `tail -f` watches live logs.

26. Set up a web server (e.g., Apache or Nginx).

Purpose: A web server delivers web pages and applications to clients through HTTP/HTTPS.

Observation: To install and configure a web server.

Implementation:

```
asif@asif-virtual-machine:~$ sudo apt install
[sudo] password for asif:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcb-1.0-0 libsbios-c2
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
asif@asif-virtual-machine:~$ sudo apt install nginx
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcb-1.0-0 libsbios-c2
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libnginx-mod-http-geoip2 libnginx-mod-http-image-filter libnginx-mod-http-xslt-filter libnginx-mod-mail libnginx-mod-stream libnginx-mod-stream-geoip2 nginx-common
Suggested packages:
  nginx-core
  fcgiwrap nginx-doc
The following NEW packages will be installed:
  libnginx-mod-http-geoip2 libnginx-mod-http-image-filter libnginx-mod-http-xslt-filter libnginx-mod-mail libnginx-mod-stream libnginx-mod-stream-geoip2 nginx nginx-common
  nginx-core
0 upgraded, 9 newly installed, 0 to remove and 0 not upgraded.
Need to get 703 kB of archives.
After this operation, 2,414 kB of additional disk space will be used.
Do you want to continue? [y/n] y
Get:1 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 nginx-common all 1.18.0-6ubuntu14.18 [40.8 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libnginx-mod-http-geoip2 amd64 1.18.0-6ubuntu14.18 [12.7 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libnginx-mod-http-image-filter amd64 1.18.0-6ubuntu14.18 [16.2 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libnginx-mod-http-xslt-filter amd64 1.18.0-6ubuntu14.18 [14.6 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libnginx-mod-mail amd64 1.18.0-6ubuntu14.18 [46.6 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libnginx-mod-stream amd64 1.18.0-6ubuntu14.18 [73.6 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libnginx-mod-stream-geoip2 amd64 1.18.0-6ubuntu14.18 [10.1 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 nginx-core amd64 1.18.0-6ubuntu14.18 [484 kB]
Get:9 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 nginx amd64 1.18.0-6ubuntu14.18 [3,884 B]
Fetched 703 kB in 2s (404 kB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 203578 files and directories currently installed.)
```

Conclusion: Nginx is a popular, fast web server for websites and reverse proxies.

27. Configure and secure a MySQL Database.

Purpose: MySQL is used to store and manage application data.

Objective: To install, configure, and secure a MySQL database server.

Implementation:

```
asif@asif-virtual-machine:~$ sudo apt install mysql-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
mysql-server is already the newest version (8.0.46-0ubuntu0.22.04.3).
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcb-1.0-0 libsbios-c2
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

Conclusion: MySQL stores structured data and `mysql_secure_installation` helps secure the server.

28. Set up a Application Server (e.g.,Apache Tomcat).

Purpose: Tomcat is a Java application server commonly used to deploy Java web applications.

Objective: To install and configure Apache Tomcat for running Java web applications.

Implementation:

```
asif@asif-virtual-machine:~$ sudo apt install apache2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap
0 upgraded, 8 newly installed, 0 to remove and 0 not upgraded.
Need to get 1,937 kB of archives.
After this operation, 7,767 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libapr1 amd64 1.7.0-8ubuntu0.22.04.2 [108 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1 amd64 1.6.1-5ubuntu4.22.04.2 [92.8 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.1-5ubuntu4.22.04.2 [11.3 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1-ldap amd64 1.6.1-5ubuntu4.22.04.2 [9,170 B]
Get:5 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-bin amd64 2.4.52-1ubuntu4.23 [1,362 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-data all 2.4.52-1ubuntu4.23 [165 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-utils amd64 2.4.52-1ubuntu4.23 [90.1 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2 amd64 2.4.52-1ubuntu4.23 [97.9 kB]
Fetched 1,937 kB in 2s (1,064 kB/s)
Selecting previously unselected package libapr1:amd64.
(Reading database ... 204092 files and directories currently installed.)
Preparing to unpack .../0-libapr1_1.7.0-8ubuntu0.22.04.2_amd64.deb ...
Unpacking libapr1:amd64 (1.7.0-8ubuntu0.22.04.2) ...
Selecting previously unselected package libaprutil1:amd64.
```

Conclusion: Tomcat runs Java-based websites and services.

29. create a service file for Apache Tomcat.(Should execute by using systemctl command)

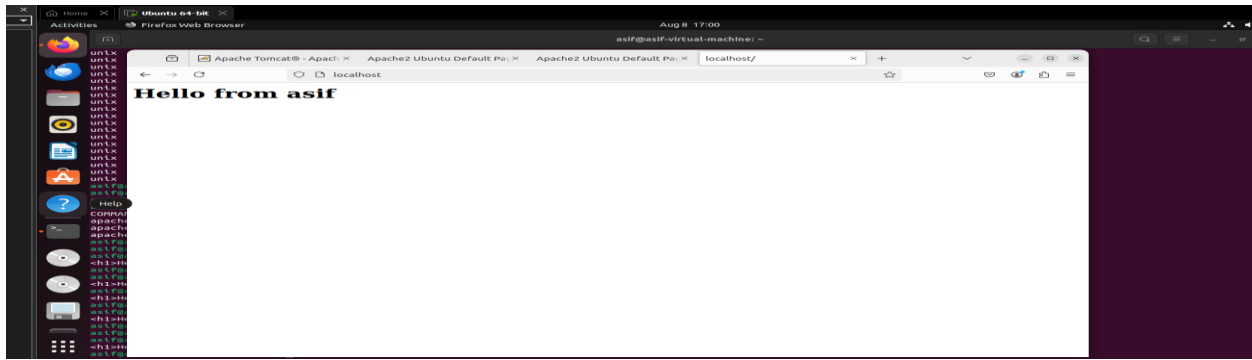
Purpose: Using `systemctl` makes it easy to start, stop, restart, enable, and monitor Tomcat.

Objective: To manage Apache Tomcat as a Linux system service.

Implementation:

```
asif@asif-virtual-machine:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
   Active: active (running) since Sat 2026-08-08 16:49:45 IST; 50s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 17334 (apache2)
  CGroup: /system.slice/apache2.service
          └─17334 /usr/sbin/apache2 -k start
            └─17336 /usr/sbin/apache2 -k start
              └─17337 /usr/sbin/apache2 -k start

Aug 08 16:49:45 asif-virtual-machine systemd[1]: Starting The Apache HTTP Server...
Aug 08 16:49:45 asif-virtual-machine apachectl[17333]: AH00558: apache2: Could not reliably det
Aug 08 16:49:45 asif-virtual-machine systemd[1]: Started The Apache HTTP Server.
```



30. Print specific columns from a delimited file.

Purpose: `awk` is useful for processing structured text such as CSV, logs, and server output.

Observation: To extract specific fields or columns from a file.

Implementation:

```
asif@asif-virtual-machine:~$ awk '{print $1, $2}' gaming
M
ihello
```

Conclusion: `awk` is useful for column-based data.

31. Filter and print lines based on a specific pattern or condition.

Purpose: Filtering is useful for finding errors, users, processes, IP addresses, or specific records.

Observation: To display only lines that match a specific pattern or condition.

Implementation:

```
asif@asif-virtual-machine:~$ printf 'hello ERROR shaik' >> shaik
asif@asif-virtual-machine:~$ echo $?
0
asif@asif-virtual-machine:~$ cat shaik
hello ERROR shaikasif@asif-virtual-machine:~$
asif@asif-virtual-machine:~$ awk 'ERROR' shaik
asif@asif-virtual-machine:~$ awk '/ERROR/' shaik
hello ERROR shaik
```

Conclusion: `awk` for column conditions.

32. Calculate and print the average, sum, or other statistics of a column.

Purpose: **awk** can calculate totals, averages, minimums, maximums, and other statistics.

Observation: To perform mathematical calculations on data stored in a file.

Implementation:

```
Rhythmbox asif@asif-virtual-machine:~$ cat hero
Alice 50
Ball 50
Doll 60
BABY 70
ASIF 90
asif@asif-virtual-machine:~$ awk '{sum += $2} END {print sum}' hero
300
asif@asif-virtual-machine:~$ awk '{sum += $2; count++} END {print sum/count}' he
ro
60
asif@asif-virtual-machine:~$ echo $?
0
```

Conclusion: **awk** is useful for sums, averages, and basic data analysis.

33. Perform string manipulation, such as extracting substrings or changing case.

Purpose: String manipulation is useful for processing usernames, logs, filenames, and application data.

Objective: To manipulate strings using Linux commands.

Implementation:

```
asif@asif-virtual-machine:~$ echo "devops" | awk '{print toupper($0)}'
DEVOPS
asif@asif-virtual-machine:~$ echo "DEVOPS" | awk '{print tolower($0)}'
devops
asif@asif-virtual-machine:~$ echo "LinuxDevOps" | awk '{print substr($0,1,5)}'
Linux
asif@asif-virtual-machine:~$ echo "Asif" | awk '{print "Hello " $0}'
Hello Asif
```

Conclusion: **awk** provides powerful functions for changing, extracting, and formatting strings.

34.Count the occurrences of a specific pattern in a file.

Purpose: This is useful for analyzing logs and finding the frequency of errors or events.

Objective: To count how many times a pattern occurs in a file.

Implementation:

```

asif@asif-virtual-machine:~$ grep -c "ERROR" shaik
1
asif@asif-virtual-machine:~$ echo $?
0
asif@asif-virtual-machine:~$ grep -ic "ERROR" shaik
1

```

Conclusion: grep combined with wc can quickly count occurrences of patterns in files.

35. Sort lines based on a specific field or column.

Purpose: Sorting helps organize data for reporting and analysis.

Objective: To arrange file records according to a specific column.

Implementation:

```

asif@asif-virtual-machine:~$ cat > employees.txt <<EOF
101,Asif,IT,25000
102,Rahul,HR,30000
103,John,Finance,35000
asif@asif-virtual-machine:~$ cat employees.txt
101,Asif,IT,25000
102,Rahul,HR,30000
103,John,Finance,35000
asif@asif-virtual-machine:~$ sort employees.txt
101,Asif,IT,25000
102,Rahul,HR,30000
103,John,Finance,35000
asif@asif-virtual-machine:~$ sort -t',' -k4,4n employees.txt
101,Asif,IT,25000
102,Rahul,HR,30000
103,John,Finance,35000
asif@asif-virtual-machine:~$ sort -t',' -k4,4nr employees.txt
103,John,Finance,35000
102,Rahul,HR,30000
101,Asif,IT,25000
asif@asif-virtual-machine:~$ sort -t',' -k4,4n employees.txt
101,Asif,IT,25000
102,Rahul,HR,30000
103,John,Finance,35000

```

Conclusion: The sort command can arrange data according to specific fields, either numerically or alphabetically.

36. Merge multiple files based on a common field or column.

Purpose: The join command is useful when two files have a common key, such as employee ID.

Objective: To combine data from multiple files using a common field.

Implementation:

```
asif@asif-virtual-machine:~$ touch file1
asif@asif-virtual-machine:~$ vi file1
asif@asif-virtual-machine:~$ vi file1
asif@asif-virtual-machine:~$ cat file1
101 Asif
102 Rahul
103 John
asif@asif-virtual-machine:~$ touch file2
asif@asif-virtual-machine:~$ vi file2
asif@asif-virtual-machine:~$ cat file2
101 IT
102 HR
103 Finance
asif@asif-virtual-machine:~$ join file1 file2
101 Asif IT
102 Rahul HR
103 John Finance
```

Conclusion: join combines related records from multiple files using a common field.

37. Substitute text in a file using search and replace.

Purpose: sed is widely used for automated text modification and configuration management.

Objective: To search for specific text and replace it with another value.

Implementation:

```
asif@asif-virtual-machine:~$ sed 's/old/new/' file1
101 Asif
102 Rahul
103 John
asif@asif-virtual-machine:~$ sed 's/old/new/g' file1
101 Asif
102 Rahul
103 John
asif@asif-virtual-machine:~$ sed -i 's/old/new/g' file1
asif@asif-virtual-machine:~$ echo $?
0
asif@asif-virtual-machine:~$ touch file
asif@asif-virtual-machine:~$ vi file
asif@asif-virtual-machine:~$ sed -i 's/Linux/DevOps/g' file
asif@asif-virtual-machine:~$ echo $?
0
asif@asif-virtual-machine:~$ cat file
Hello DevOps
Hello DevOps
```

Conclusion: sed provides a simple and powerful way to automate search-and-replace operations.

38. Delete specific lines based on a pattern or line number.

Purpose: Deleting unwanted records is useful when cleaning configuration files, logs, or datasets.

Objective: To remove unwanted lines from a file.

Implementation:

```
asif@asif-virtual-machine:~$ touch keys
asif@asif-virtual-machine:~$ vi keys
asif@asif-virtual-machine:~$ cat keys
Help
error
DevOps
success
asif@asif-virtual-machine:~$ sed -i '/error/d' keys
asif@asif-virtual-machine:~$ cat keys
Linux
DevOps
success
```

Conclusion: sed can delete lines by line number or by matching a specific pattern.

39. Append or insert text before or after a specific pattern or line.

Purpose: This is useful for automatically modifying configuration files.

Objective: To insert or append text into a file.

Implementation:

```
asif@asif-virtual-machine:~$ sed -i '/error/d' keys
asif@asif-virtual-machine:~$ cat keys
Help
DevOps
success
asif@asif-virtual-machine:~$ sed -i '/Linux/a ubuntu' keys
asif@asif-virtual-machine:~$ cat keys
Linux
ubuntu
DevOps
success
```

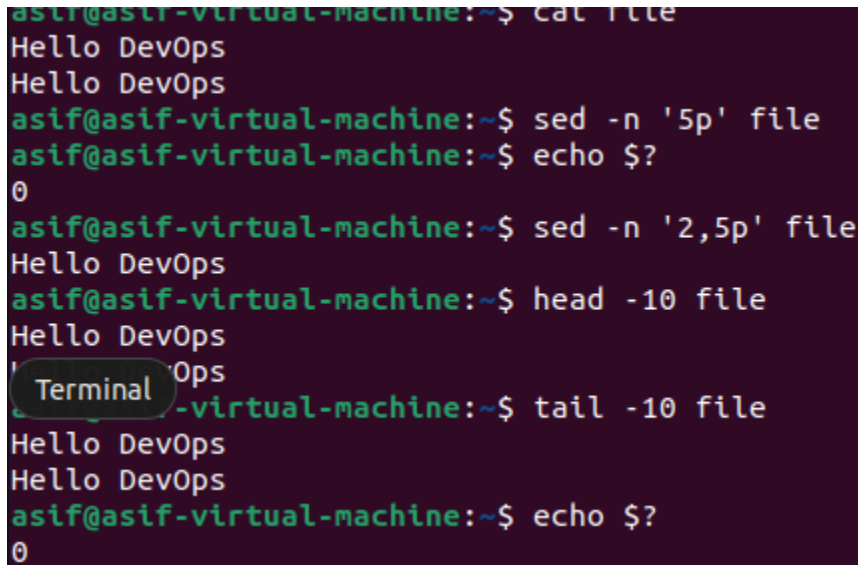
Conclusion: sed allows text to be inserted before or after specific lines or patterns.

40. Print only specific lines from a file.

Purpose: This helps administrators inspect specific portions of large files.

Objective: To display only selected lines from a file.

Implementation:

A terminal window with a dark purple background and green text. The prompt is 'asif@asif-virtual-machine:~\$'. The user enters 'cat file', and the output is 'Hello DevOps' on two lines. Then, the user enters 'sed -n '5p' file', and the output is '0'. Next, the user enters 'sed -n '2,5p' file', and the output is 'Hello DevOps' on two lines. Then, the user enters 'head -10 file', and the output is 'Hello DevOps' on two lines. A 'Terminal' label is overlaid on the left. Finally, the user enters 'tail -10 file', and the output is 'Hello DevOps' on two lines. The user then enters 'echo \$?', and the output is '0'.

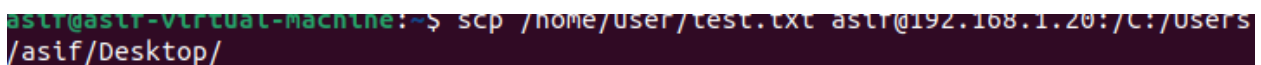
Conclusion: Commands such as `sed`, `head`, and `tail` make it easy to display only the required portion of a file.

41. Copy file from linux to windows machine

Purpose: File transfer is commonly required in DevOps for moving scripts, logs, configuration files, and application packages.

Objective: To transfer files from a Linux machine to a Windows machine.

Implementation:

A terminal window with a dark purple background and green text. The prompt is 'asif@asif-virtual-machine:~\$'. The user enters 'scp /home/user/test.txt asif@192.168.1.20:/C:/Users/asif/Desktop/'.

Conclusion: `scp`/SFTP provides secure file transfer between Linux and Windows systems.

42. 5 use cases for AWK and 5 use cases for sed

Purpose: Extract a particular field.

```
asif@asif-virtual-machine:~$ awk -F',' '{print $2}' employees.txt
Asif
Rahul
John
asif@asif-virtual-machine:~$ echo $?
```

Objective: Print specific columns

Purpose: Display only employees belonging to IT.

Objective: Filter records

Implementation:

```
asif@asif-virtual-machine:~$ awk -F',' '$3=="IT" {print $0}' employees.txt
101,Asif,IT,25000
```

Purpose: Calculate total salary.

Objective: Calculate sum

Implementation:

```
asif@asif-virtual-machine:~$ awk -F',' '{sum += $4} END {print sum}' employees.txt
90000
```

Purpose: Calculate average salary.

Objective: Calculate average

Implementation:

```
asif@asif-virtual-machine:~$ awk -F',' '{sum += $4; count++} END {print sum/count}' employees.txt
30000
```

Purpose: Convert strings to uppercase

Objective: String manipulation

Implementation:

```
asif@asif-virtual-machine:~$ echo "devops" | awk '{print toupper($0)}'
DEVOPS
```

Awk Conclusion: awk is mainly used for column-based processing, filtering, calculations, reporting, and text manipulation.

Purpose: Replace text.

Objective: Search and replace

Implementation:

```
asif@asif-virtual-machine:~$ sed 's/Linux/DevOps/g' file
Hello DevOps
Hello DevOps
```

Purpose: Delete line number 2.

Objective: Delete a line

Implementation:

```
asif@asif-virtual-machine:~$ sed '2d' file
Hello DevOps
```

Purpose: Remove lines containing error.

Objective: Delete lines matching a pattern

Implementation:

```
asif@asif-virtual-machine:~$ sed '/error/d' file
Hello DevOps
Hello DevOps
```

Purpose: Insert text before line 2.

Objective: Insert text

Implementation:

```
asif@asif-virtual-machine:~$ sed '2i New Line' file
Hello DevOps
New Line
Hello DevOps
```

Purpose: Print lines 5 to 10.

Objective: Print selected lines

Implementation:

```
asif@asif-virtual-machine:~$ sed -n '5,10p' file
asif@asif-virtual-machine:~$ echo $?
0
```

Conclusion: sed is mainly used for searching, replacing, deleting, inserting, and modifying text.